MASTER RESEARCH PROPOSAL

The Quantum Cryptanalytic Threat Landscape and Post-Quantum Migration: An Analytical Survey of Hardware Realities, Standards, and Continuous Risk Frameworks

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ABSTRACT

The rapid advancement of quantum computing presents a fundamental challenge to classical cryptographic systems, necessitating a comprehensive understanding of both algorithmic and hardware progress. This survey research proposal systematically examines the evolving quantum threat landscape along three interconnected dimensions. First, it reviews the evolution of quantum cryptanalytic algorithms, from foundational results such as Shor's algorithm and Grover's algorithm to more recent refinements and alternative constructions, including Regev-type frameworks, highlighting how these developments affect both asymmetric cryptosystems and the effective security of symmetric primitives. Second, it analyses recent reductions in quantum resource requirements, with particular emphasis on fault-tolerant architectures and quantum error-correction techniques that significantly lower physical qubit overhead under optimistic assumptions. Third, it evaluates progress in quantum hardware, including improvements in qubit coherence, noise resilience, and material platforms, which collectively influence the feasibility and timeline of large-scale fault-tolerant quantum computation.

In parallel, the paper reviews the status of post-quantum cryptographic standardisation, focusing on National Institute of Standards and Technology FIPS 203–205 and ongoing developments, and discusses implementation-level vulnerabilities to highlight deployment challenges. A risk assessment framework based on Mosca's inequality is applied to contextualise these advances under the "harvest now, decrypt later" threat model. The analysis demonstrates that cryptographic risk is shaped not only by algorithmic breakthroughs but also by reductions in qubit overhead and improvements in hardware stability, both of which compress the timeline to cryptanalytically relevant quantum capabilities. Based on these findings, the paper provides practical recommendations to support cryptographic agility and a secure transition to post-quantum standards.

CHAPTER 1: INTRODUCTION AND RESEARCH FOUNDATION

1.1 Background and Context

Modern digital security is foundational to global communication and commerce, relying upon a combination of asymmetric and symmetric cryptographic primitives. Asymmetric systems, such as RSA and Elliptic Curve Cryptography (ECC), facilitate essential services including secure key exchange, digital signatures, and identity verification. These schemes derive their security from the computational hardness of integer factorization and the discrete logarithm problem (DLP)—mathematical challenges that are efficiently solvable via Shor's quantum algorithm. Consequently, classical public-key infrastructures are entirely vulnerable to the eventual realization of a cryptanalytically relevant quantum computer (CRQC; also referred to as a cryptographically relevant quantum computer). Symmetric primitives, including AES and SHA-2, are not broken outright by quantum algorithms; however, Grover's algorithm provides a generic quadratic speed-up for brute-force search and thus reduces their effective security margins, motivating the use of larger key and digest sizes (e.g., AES-256, SHA-384/512) for post-quantum robustness.

The timeline for these threats is increasingly shaped by a global "fault-tolerance race", characterized by divergent architectural approaches. Google and IBM have primarily focused on superconducting transmon qubits, while Microsoft pursues a qualitatively different path based on topological qubits, with recent claims of a quantum-computing breakthrough, albeit accompanied by scientific skepticism. Recent peer-reviewed theoretical approaches, including logical operator gauging, suggest the potential for quantum low-density parity-check (qLDPC) codes to reduce error-correction overhead. In parallel, experimental efforts such as IBM's prototype processors explore feasible hardware architectures; however, scalable fault-tolerant performance has not yet been demonstrated.

In response, NIST released the first generation of standardized post-quantum algorithms (FIPS~203/204/205) and announced a forthcoming signature standard (FIPS~206) to strengthen long-term security postures.

Recent advances in fault-tolerant architectures, neutral-atom systems, and quantum networking have led some researchers to argue that useful large-scale quantum systems may arrive substantially earlier than previously anticipated. Researchers affiliated with Harvard's Quantum Initiative suggested that certain fault-tolerant capabilities could emerge "by the end of this decade," reflecting growing momentum in both academia and industry.

This paper contributes a consolidated assessment of the modern quantum threat landscape by integrating developments in quantum algorithms, fault-tolerant quantum architectures, and post-quantum standardization efforts into a unified analytical framework. In addition to surveying the state of the art, the paper contextualizes these developments through Mosca-based risk analysis and discusses their implications for cryptographic migration planning and long-term security policy.

1.2 Research Motivation

A CRQC would break widely deployed public-key systems based on integer factorization and discrete logarithms, endangering confidentiality and authenticity of internet-scale systems. Even before such a device exists, adversaries can mount "*harvest now decrypt later*" (HNDL) attacks by storing encrypted traffic today for future decryption—elevating the urgency of PQC migration for data with long confidentiality lifetimes.

1.3 Research Questions

- **RQ1**: How do recent advances in quantum algorithms, fault-tolerant architectures, and quantum hardware affect the projected feasibility and timeline of CRQCs?
- **RQ2**: What are the implications of Shor's and Grover's algorithms for the long-term security of contemporary asymmetric and symmetric cryptographic systems?
- **RQ3**: How effective are current National Institute of Standards and Technology (NIST) standardized post-quantum cryptographic schemes in mitigating emerging quantum threats, and what implementation challenges remain?
- **RQ4**: What limitations exist in traditional interpretations of Mosca's Inequality for modeling HNDL threats?
- **RQ5**: How can continuous-time information-value profile models improve the understanding of long-term HNDL exposure and post-quantum migration risk?

1.4 Purpose Statement

The purpose of this analytical survey is to synthesize current research on quantum cryptanalysis, fault-tolerant quantum computing, post-quantum cryptographic standardization, and HNDL threat modeling in order to evaluate the evolving quantum threat landscape and its implications for long-term cryptographic security. The study further aims to conceptually examine continuous-time risk interpretations of Mosca's Inequality and

provide practical recommendations to support cryptographic agility and post-quantum migration planning.

1.5 Research Objectives

- To analyze how recent advances in quantum algorithms, fault-tolerant architectures, and quantum hardware influence the projected feasibility and timeline of CRQCs.
- To evaluate the implications of Shor's and Grover's algorithms for the long-term security of contemporary asymmetric and symmetric cryptographic systems.
- To examine the effectiveness, standardization status, and implementation challenges of current NIST-standardized and emerging post-quantum cryptographic schemes.
- To investigate the limitations of traditional interpretations of Mosca's Inequality in modeling HNDL threats.
- To conceptually analyze how continuous-time information-value profile models may improve the understanding of long-term HNDL exposure and post-quantum migration risk.

1.6 Organization of Research

This proposal has been organized in the following chapters.

Chapter	Contents
Chapter 1	Introduction, research problem, objectives.
Chapter 2	Literature review and theoretical background
Chapter 3	Research methodology
Chapter 4	Proposed Framework
Chapter 5	Challenges, Conclusion
References	Citations and links used in this research
Appendices	Mathematical model derivation with examples

CHAPTER 2: LITERATURE REVIEW AND THEORETICAL BACKGROUND

This chapter gives a brief introduction to quantum computing principles and foundations to clear confusion between classical and quantum computers.

Then it incorporates peer-reviewed journal articles, conference proceedings, NIST PQC standards, and technical reports. It also consolidates selected industry publications related to quantum cryptanalysis, PQC, fault-tolerant quantum computing, and HNDL threat modeling. Foundational references were selected based on their relevance to established quantum algorithms, including Shor's and Grover's algorithms, as well as contemporary developments in lattice-based cryptography, hash-based signatures, quantum error correction, and CRQC resource estimation.

In addition to formal academic literature, selected industry and institutional reports were included where appropriate to reflect rapidly evolving developments in quantum hardware, migration planning, and operational cybersecurity risk. Such sources were treated as supplementary evidence rather than definitive scientific consensus and were interpreted conservatively within the broader context of peer-reviewed research.

2.1 Foundations of Quantum Computing

Quantum computing is founded upon principles of quantum mechanics that differ fundamentally from classical physics. While classical computers process information using bits that exist in one of two states, 0 or 1, quantum computers employ quantum bits (**qubits**), which can exist in a superposition of multiple states simultaneously. The computational power of quantum systems arises from several uniquely quantum phenomena, including superposition, entanglement, and interference.

The conceptual foundations of quantum computation were established by Feynman, who proposed the use of quantum systems to efficiently simulate physical processes that are intractable for classical computers [1]. Deutsch subsequently formalized the concept of a universal quantum computer capable of performing arbitrary quantum computations [2]. These pioneering works established the theoretical framework for the development of quantum algorithms that would later demonstrate significant computational advantages over classical approaches.

2.1.1 Quantum Superposition

Superposition allows a qubit to represent a linear combination of basis states, enabling a quantum computer to process multiple computational paths concurrently.

2.1.2 Quantum Entanglement

Entanglement establishes non-classical correlations between qubits, allowing information to be distributed across a quantum system in ways that cannot be replicated classically. Quantum interference enables quantum algorithms to amplify the probabilities of correct outcomes while suppressing incorrect ones, thereby producing computational advantages for specific classes of problems [3], [4].

2.1.3 Quantum Circuits

A quantum circuit is the quantum computing equivalent of a classical logic circuit, designed to perform computations using qubits. Quantum circuits consist of three main stages: initialization, processing, and readout. During initialization, qubits are set to a known state, typically all 0s. Processing involves applying quantum gates to transform qubit states, and readout measures the qubits to produce classical results for practical use.

2.2 Historical Development of Quantum Algorithms

The evolution of quantum computing has been marked by a series of theoretical and experimental breakthroughs. Early work focused on establishing the computational model and identifying problems for which quantum systems could outperform classical computers. Significant milestones include the development of the Deutsch--Jozsa algorithm, which demonstrated an early example of quantum computational advantage, followed by the discovery of Shor's and Grover's algorithms, which revealed profound implications for modern cryptography [5],Shor1994,Grover1996}.

The progression of quantum computing research is summarized in Table 2.1.

Table 2.1: Selected milestones in the development of quantum computing.

1980-1982	Yuri Manin and Feynman propose quantum simulation as a motivation for quantum computers [1].
1985	Deutsch introduces the universal quantum computer model [2].
1992	Deutsch--Jozsa algorithm demonstrates quantum computational advantage [5].

1994	Shor develops a polynomial-time quantum algorithm for integer factorization and discrete logarithms [6].
1996	Grover introduces a quadratic-speedup quantum search algorithm [7].
2001	First experimental demonstration of Shor's algorithm on a small-scale quantum processor [8].
2019	Google reports a quantum supremacy experiment using a superconducting quantum processor [9].
2024-2026	Significant advances in fault-tolerant architectures, qLDPC codes, and quantum resource estimation accelerate progress toward cryptographically relevant quantum computers.

2.3 Quantum Algorithms and Cryptographic Implications

The security implications of quantum computing became evident with the development of Shor's algorithm, which demonstrated that integer factorization and discrete logarithm problems can be solved in polynomial time on a sufficiently large fault-tolerant quantum computer [6]. Since the security of widely deployed public-key cryptographic systems, including RSA, Diffie--Hellman, and Elliptic Curve Cryptography (ECC), relies on the computational difficulty of these problems, Shor's algorithm presents a direct threat to contemporary public-key infrastructures.

In contrast, Grover's algorithm provides a quadratic speedup for unstructured search problems [7]. Rather than completely breaking symmetric cryptographic systems, Grover's algorithm effectively reduces the security strength of symmetric keys by approximately one-half. Consequently, cryptographic standards often recommend increasing key sizes, such as transitioning from AES-128 to AES-256, to maintain equivalent security margins against quantum adversaries [10].

Together, Shor's and Grover's algorithms fundamentally altered the long-term security assumptions underlying modern cryptography and motivated the development of post-quantum cryptography (PQC). While Shor's algorithm threatens public-key cryptography through exponential reductions in computational complexity, Grover's algorithm necessitates adjustments to symmetric-key security parameters. These developments have driven international efforts, including the National Institute of Standards and Technology (NIST) Post-Quantum Cryptography Standardization Project, to develop and standardize cryptographic algorithms resistant to both classical and quantum attacks [11].

2.3.1 Shor-type Factoring and Resource Estimates

Shor's algorithm [12] provides a polynomial-time quantum method for factoring and discrete logarithms, with asymptotic complexity on the order of $\tilde{O}(k^2)$ to $\tilde{O}(k^3)$ with optimized arithmetic) for a k-bit modulus. Under optimistic assumptions about fault-tolerant quantum hardware—particularly logical gate speeds in the MHz range and efficient magic-state distillation—these scalings suggest that factoring an RSA-2048 modulus could, in principle, be achieved in hours to days.

Concrete resource estimates, however, remain highly sensitive to error correction schemes and architectural constraints.

Early surface-code analyses (e.g., by Craig Gidney and Martin Ekerå) indicated requirements on the order of ~20 million physical qubits for sub-day runtimes. More recent refinements, incorporating improved compilation and distillation strategies, reduce this to below one million physical qubits under favorable assumptions, with runtimes ranging from hours to under a week. [13]

Alternative theoretical approaches based on qLDPC codes aim to substantially reduce qubit overhead. For example, the "Pinnacle" architecture [14] proposes a fault-tolerant design in which RSA-2048 factoring could be achieved with fewer than $\sim 10^5$ physical qubits under specific hardware assumptions, including physical error rates on the order of 10^{-3} and microsecond-scale error-correction cycles.

Recent work by Cain et al. (2026) demonstrated that cryptographically relevant implementations of Shor's algorithm may be achievable using as few as 10,000 reconfigurable atomic qubits through the combination of high-rate qLDPC codes, reconfigurable neutral-atom architectures, and optimized logical operations [15].

However, these estimates rely on tightly integrated architectural features—such as high-throughput magic state distillation ("magic engines"), modular qLDPC processing units, and efficient classical feedforward—that are not yet experimentally demonstrated at scale. As a result, while such proposals indicate a plausible path to order-of-magnitude reductions in qubit counts relative to surface-code-based designs, they remain significantly more speculative and hardware-sensitive, often involving more demanding implementation

requirements and potentially longer runtimes. Although substantial engineering challenges remain, these results further illustrate how architectural innovation can significantly compress projected CRQC resource requirements and contribute to uncertainty in quantum-threat timelines.

In contrast, a controversial 2026 proposal by Tim Palmer argues that standard quantum mechanics may overestimate the scalability of entanglement and quantum computation, predicting that quantum computers could encounter a fundamental physical ceiling on the order of a few hundred qubits. This prediction remains highly speculative and is not accepted by the mainstream quantum-computing community [16].

Overall, while the asymptotic advantage of Shor's algorithm is well established, practical timelines for breaking RSA-2048 depend critically on advances in fault-tolerant architectures, error correction, and large-scale quantum hardware.

2.3.2 Regev's Efficient Quantum Factoring and Elliptic Curves

Regev introduced a factoring framework with a lower gate complexity per run but more repetitions under a heuristic number-theoretic assumption, with analyses exploring space/noise trade-offs and practical considerations. [17] Subsequently, Barbulescu, Barbulescu, Barcau, and Paşol extended this framework to elliptic-curve discrete logarithms (ECDLP), identifying parameter regimes where it may outperform Shor asymptotically and describing instances of failure/success under specific heuristics. [18]

2.3.3 Grover's Algorithm and Symmetric Primitives

Grover's algorithm yields a quadratic speed-up for unstructured search; in cryptography this implies that idealized key search over a k-bit key space requires roughly $2^{\frac{k}{2}}$ evaluations. [7], [13] This reduction affects all symmetric primitives, including block ciphers, MACs, and hash functions. However, unlike Shor's algorithm for RSA and ECC—which completely breaks those schemes—a quadratic speed-up merely reduces effective security, rather than eliminating it outright.

While no known algorithm asymptotically surpasses Grover's bound for unstructured symmetric-key search, ongoing advances in hybrid quantum optimization, oracle engineering, circuit reduction, and variational quantum algorithms may progressively reduce the practical resource overheads associated with future quantum computational and cryptanalytic implementations. [20]

2.4 Guidance and Standards on Symmetric Security in the Quantum Era

Government agencies and international organizations have established roadmaps for transitioning to quantum-resilient cryptography. Key initiatives from NIST, the NSA, and the European Union define standards, migration priorities, and implementation timelines, although their scope and requirements differ.

2.4.1 NIST

NIST key-management recommendations and cost analyses provide baseline guidance for selecting symmetric-key parameters suitable for long-lived

data in the presence of quantum adversaries, including the use of AES-256 and SHA-384/512 to retain adequate post-quantum security margins. [21]

While existing literature extensively characterizes the algorithmic capabilities of quantum adversaries, the practical realization of such threats is fundamentally constrained by the underlying quantum hardware. Accordingly, a system-level understanding of current quantum computing platforms is necessary to contextualize cryptanalytic feasibility.

Recent surveys have examined enterprise migration risks associated with CRQCs, including HNDL threats, sector-specific vulnerabilities, and post-quantum cryptographic transition challenges. [22] However, such analyses generally model confidentiality exposure using fixed secrecy lifetimes or discrete migration horizons. In contrast, the present survey additionally considers continuously decaying information-value models and rolling HNDL exposure dynamics, in which residual confidentiality value may persist beyond nominal secrecy deadlines.

2.5 Quantum Hardware Landscape and Vendor Comparison

The realization of CRQC is constrained not only by algorithmic advances but also by the physical characteristics of qubit implementations. As of 2026, no dominant hardware paradigm has emerged; instead, multiple competing qubit modalities coexist, each with distinct trade-offs in coherence, temperature requirements, scalability, and error susceptibility. These hardware-level considerations play a decisive role in determining the feasibility and timeline of large-scale quantum computation.

2.5.1 Major Industrial Approaches

Leading technology vendors have adopted fundamentally different qubit strategies [23], [24], [25]:

- **IBM / Google**: Superconducting transmon qubits based on Josephson junction circuits.
- **Intel**: Silicon spin qubits implemented in semiconductor quantum dots.
- **Microsoft**: Topological qubits based on Majorana quasiparticles (currently experimental).

These approaches differ in physical realization, which directly influences operational constraints, coherence properties, and scalability pathways.

2.5.2 Comparative Hardware Characteristics

Table 2.1:Comparison of major quantum computing hardware platforms as of 2026. Reported values are representative approximate estimates and may vary across implementations and experimental conditions.

Attribute	IBM / Google	Intel	Microsoft
Qubit Type	Transmon	Spin (Si)	Topological
Physical Platform	Superconducting circuits	Quantum dots	Majorana modes
Operating Temperature	~10–20 mK	~50–100 mK	~20 mK (target)
Coherence Time	~50–200 µs	µs–ms	Theoretical (long)
Gate Speed	~10–100 ns	ns–µs	No validated value
Scalability	100–1000+ qubits	Early-stage	Experimental
Key Limitation	Decoherence, cooling	Variability	Unproven physics

As shown in Table 2.1, superconducting platforms currently dominate in scale and maturity, while silicon spin qubits offer improved thermal tolerance and potential compatibility with existing semiconductor manufacturing. Topological approaches aim to address decoherence at a fundamental level but remain experimentally unproven. Consequently, hardware modality—not merely qubit count—plays a central role in assessing CRQC feasibility.

Recent proposals for three-dimensional self-correcting quantum memories suggest that quantum information may eventually be preserved for long durations with substantially reduced active error-correction overhead. If scalable, such approaches could significantly lower the resource requirements associated with fault-tolerant quantum computation, thereby accelerating the timeline toward CRQCs. However, fully practical self-correcting quantum memory in three dimensions remains an open research challenge. [26]

2.5.3 Emerging Hardware Trends

Recent research reveals several converging trends across quantum hardware platforms [9], [14]:

2.5.3.1 Relaxation of Cryogenic Constraints

Traditional superconducting systems require ultra-low temperatures (~10--20 mK). Emerging qubit platforms demonstrate operation at elevated temperatures (≥ 100 mK), reducing reliance on extreme cryogenic infrastructure and simplifying system engineering.

2.5.3.2 Noise-Resilient Qubit Design

The field is transitioning from minimizing environmental noise to engineering systems that are intrinsically robust to noise. This includes advances in materials, fabrication processes, and qubit architectures designed to tolerate non-ideal operating conditions.

2.5.3.3 Material-Driven Quantum Computing

Materials science is increasingly central to quantum computing progress. The choice of substrate directly influences coherence time, noise susceptibility, and scalability. Advances in superconducting materials, silicon purification, and alternative hosts are driving improvements across platforms.

2.5.3.4 Coherence--Scalability Trade-off

A persistent trade-off exists between coherence and scalability. Superconducting systems are highly scalable but exhibit relatively short coherence times, whereas spin-based and atomic systems often achieve longer coherence but present challenges in large-scale integration.

2.5.4 Impact of Recent Advances

Recent work on electron qubits hosted in solid neon introduces a new design axis characterized by higher operating temperatures (~100-400 mK) and improved noise resilience. [28] While coherence times remain on the order of microseconds, these systems demonstrate that qubit stability may be achieved under less restrictive thermal conditions.

2.5.4.1 Implications

- **Reduced Thermal Constraints**: Higher operating temperatures alleviate the engineering burden associated with dilution refrigeration.
- **Alternative Scaling Pathways**: Novel material platforms suggest that future quantum systems may not rely exclusively on superconducting architectures.
- **Increased Timeline Uncertainty**: Hardware breakthroughs introduce variability in projections of CRQC realization timelines.

2.5.5 Relevance to Cryptanalytic Risk

The diversity of hardware approaches introduces structural uncertainty into quantum threat modelling. Progress is inherently non-linear and architecture-dependent, with potential breakthroughs arising from unexpected physical implementations. Hardware advances may reduce error-correction overhead, increase feasible qubit counts, and shorten execution timelines.

Accordingly, the timeline to cryptographically relevant quantum computation is not solely a function of algorithmic efficiency but is increasingly determined by advances in materials science and quantum hardware engineering.

CHAPTER 3: RESEARCH METHODOLOGY

This study adopts an analytical survey methodology based on literature synthesis, comparative analysis, and conceptual risk modeling. The research integrates peer-reviewed journal articles, conference papers, NIST standards publications, technical reports, and selected industry sources related to quantum cryptanalysis, fault-tolerant quantum computing, post-quantum cryptography (PQC), and HNDL threat modeling.

The study first reviews foundational and contemporary quantum algorithms, including Shor's and Grover's algorithms, together with recent developments in quantum resource estimation, quantum low density parity check (qLDPC) based fault tolerance, and quantum hardware engineering. It then comparatively examines NIST-standardized PQC schemes and associated implementation-level security considerations.

To analyze long-term cryptographic exposure, the research employs a conceptual analytical framework based on Mosca's Inequality and extends it using continuous-time information-value decay models. Discrete and continuous convolution-based formulations are used to illustrate the temporal accumulation and decay of HNDL exposure under varying secrecy persistence assumptions.

Finally, the study synthesizes the reviewed literature and analytical interpretations to derive practical migration recommendations and identify future research directions related to post-quantum cybersecurity and cryptographic resilience

APPENDICES

APPENDIX A: Hypothesized HNDL Value Accumulation Model

To study the long-term implications of HNDL attacks under the emergence of CRQCs, a generalized mathematical model is first constructed to describe the temporal accumulation and decay of harvested encrypted information value.

The proposed formulation begins from a discrete-time perspective and is later extended toward a continuous-time representation, ultimately revealing a convolution-equivalent structure.

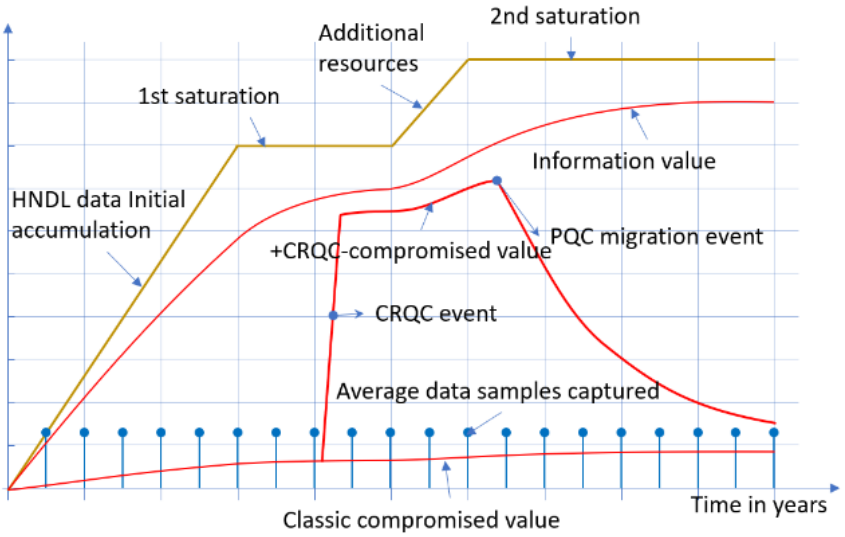


Figure 1: General HNDL profile

The model is constructed under the following assumptions:

- Time is discretized into ordered intervals:

$$t_i = i\Delta t \tag{1}$$

where Δt denotes the sampling interval.

- At each interval t_i, a quantity of encrypted information $D[i]$ is harvested and stored by an adversary.
- The informational value of harvested data decreases over time due to:

- expiration of secrecy relevance,
- operational obsolescence,
- credential rotation,
- information aging.

- The adversary may later obtain quantum capabilities capable of breaking vulnerable public-key cryptography.
- Post-Quantum Cryptography (PQC) migration may eventually terminate future quantum-vulnerable exposure while leaving previously harvested ciphertext potentially vulnerable.

APPENDIX B: Discrete-Time Information Value Model

Let $D[i]$ denote the quantity or informational significance of data harvested at interval j.

The value of this harvested information at a later interval i depends on a temporal weighting function describing secrecy persistence.

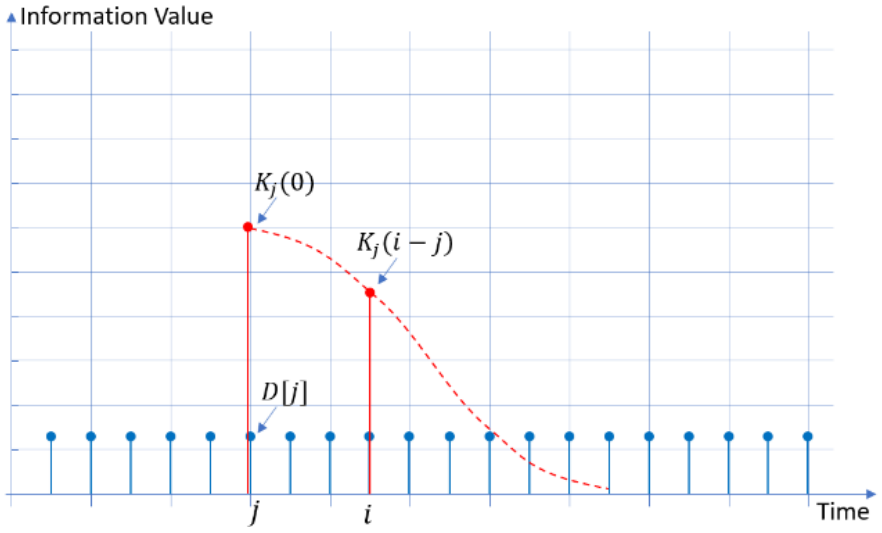


Figure 2: Sample profile for derivation

A generalized temporal decay kernel is defined as:

$$K(i-j) \tag{2}$$

where:

- $K(0) = 1$,
- $K(x)$ monotonically decreases with increasing age x,

- $K(x) \to 0$ for sufficiently large x.

The instantaneous value contribution of sample $D[j]$ observed at time i becomes:

$$V_j[i] = D[j]K(i-j) \tag{3}$$

The total virtual value accumulated at interval i is therefore obtained by summing all surviving historical contributions:

$$V_v[i] = \sum_{j=0}^{i} D[j]K(i-j) \tag{4}$$

This equation represents the total residual informational value available to an adversary at time i. Mathematically, it may be interpreted as a causal finite-length discrete convolution between the harvested data sequence $D[j]$ and the secrecy persistence kernel $K(i-j)$.

B.1 Example Decay Profiles

Exponential Decay Profile: A continuously decaying secrecy profile may be modeled as:

$$K(i-j) = e^{-\frac{i-j}{\tau}} \tag{5}$$

where τ denotes the secrecy decay time constant.

The resulting value model becomes:

$$V_v[i] = \sum_{j=0}^{i} D[j]\, e^{-\frac{i-j}{\tau}} \tag{6}$$

This profile models gradually aging information such as:

- strategic intelligence,
- credentials,
- tactical metadata,
- encrypted communications.

Finite-Lifespan Secrecy Profile: Certain information may instead retain value for a finite duration before abruptly becoming irrelevant.

This can be modeled using a rectangular persistence kernel:

$$K(i-j) = \Pi\left(\frac{i-j}{L}\right) \tag{7}$$

where:

$$\Pi(\mathrm{x}) = \begin{cases} 1, & 0 \le x < 1 \\ 0, & \text{otherwise} \end{cases}$$

and L denotes the secrecy lifespan.

The corresponding virtual value becomes:

$$V_v[i] = \sum_{j=0}^{i} D[j]\,\Pi\left(\frac{i-j}{L}\right) \tag{8}$$

This profile models:

- temporary operational secrets,
- session credentials,
- short-lived intelligence,
- expiring cryptographic assets.

APPENDIX C: Transition Toward Continuous-Time Modeling

The discrete formulation may be generalized toward continuous-time representations.

Consider each harvested sample as a rectangular information pulse of duration Δt:

$$D[j] \to D(\tau)\Delta\tau \tag{9}$$

The accumulated virtual value then becomes:

$$V_v(t) \approx \sum_{j} D(\tau)K(t-\tau)\Delta\tau \tag{10}$$

As the sampling interval approaches infinitesimal resolution, $\Delta\tau \to d\tau \to 0$, the discrete summation converges toward the Riemann integral:

$$V_v(t) = \int_0^t D(\tau)K(t-\tau)d\tau \tag{11}$$

This expression corresponds to the convolution:

$$V_v(t) = (D * K)(t) \tag{12}$$

which reveals that the total residual information value is equivalent to the convolution of harvested data generation with a secrecy persistence kernel.

Continuous Exponential Decay Example: The exponential secrecy persistence term correspondingly becomes:

$$e^{-\left(\frac{i-j}{\tau}\right)} \to e^{-\left(\frac{t-\tau}{T_c}\right)} \tag{13}$$

where T_c denotes the continuous-time secrecy decay constant.

This yields:

$$V_v(t) = \int_0^t D(\tau) e^{-\left(\frac{t-\tau}{T_c}\right)} d\tau \tag{14}$$

which represents the continuous-time exponential secrecy accumulation model.

C.1: Interpretation

The convolution formulation provides several important interpretations:

- The harvested data function $D(t)$ describes adversarial collection activity.
- The kernel $K(t)$ describes secrecy persistence dynamics.
- The resulting convolution quantifies the time-varying exploitable informational value available to an adversary.
- Different secrecy models correspond to different kernel selections.
- PQC migration may be modeled as truncation or modification of the quantum-vulnerable portion of the convolution integral.

The resulting framework provides a generalized mathematical basis for analyzing HNDL persistence, CRQC timing impact, secrecy decay behavior, and residual post-migration exposure.

APPENDIX D: Derivation of Observable Compromise Value

We reconstruct the observable compromise model starting from the discrete-time accumulation framework and extending it to CRQC and classical adversarial effects.

Let $V_o[i]$ denote the observable information value accessible to an adversary at time i, defined as the convolutional accumulation of harvested data:

$$V_o[i] = \sum_{j=0}^{i} D[j] K(i-j) \tag{15}$$

where $D[j]$ is the harvested data sequence and $K(\cdot)$ is the secrecy persistence kernel.

D.1: Pre-CRQC / Classical Regime

In the absence of quantum capabilities, only a small fraction of the observable value becomes exploitable:

$$V_c[i] = \eta_c V_o[i], \quad 0 < \eta_c \ll 1 \tag{16}$$

This represents classical compromise through traditional cryptanalytic or implementation-based attacks.

D.2: CRQC-Enabled Compromise

We now consider the case where a CRQC event occurs at time t_{CRQC}, and PQC migration occurs at time $t_{\mathrm{PQC}} = l$.

For all information collected prior to PQC migration, the adversary retains a residual observable state $V_o[l]$.

When quantum capability becomes available, the adversary can retrospectively decrypt a fraction of this stored state:

$$V_q[i] = \eta_q V_o[i], i \geq l \tag{17}$$

where η_q represents the quantum compromise efficiency.

D.3: Total Observable Compromise

The total adversarially accessible value is therefore piecewise defined as:

$$V_o^{\mathrm{Comp}}[i] = V_c[i] + V_q[i] = \eta_c V_o[i] + \begin{cases} 0 & i < t_{\mathrm{CRQC}} \\ \eta_q V_o[i], & t_{\mathrm{CRQC}} < i < l, \\ \eta_q V_o[l], & i \geq l \end{cases} \tag{18}$$

D.4: Interpretation

This formulation highlights two distinct mechanisms of compromise:

- **Classical leakage**, which continuously scales with the instantaneous observable value $V_o[i]$.
- **Quantum retrospective compromise**, which locks onto the frozen state $V_o[l]$ at the moment of PQC migration and enables post-hoc decryption.

APPENDIX E: Numerical Modeling and Visualization

Approximate numerical models were implemented in Python to illustrate representative HNDL exposure dynamics under different confidentiality assumptions. The modeled confidentiality-value function, denoted as $V_h(t)$, was evaluated using two primary scenarios:

1. Fixed-lifetime secrecy models, in which information retains constant value until a specified expiration horizon.
2. Exponentially decaying secrecy models, in which confidentiality value gradually decreases as a function of time.

For representative exponentially decaying scenarios, the confidentiality-value model may be expressed as:

$$V_h(t) = V_0\, e^{-\lambda t}$$

where V_0 denotes the initial confidentiality value and λ represents a decay parameter governing the rate of information-value reduction over time.

The resulting simulations and figures were generated to visualize comparative HNDL exposure behavior under varying assumptions regarding secrecy persistence, migration delays, and evolving adversarial capabilities. These models are intended primarily for conceptual illustration and comparative interpretation rather than quantitative prediction of real-world cryptographic compromise timelines.

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